

Fault scenario	Mho Zone 1	Quad Zone 1	Mho Zone 2	Quad Zone 2	87L
3PH metallic, $k = 0.15$, α	TRIP	TRIP	TRIP	TRIP	TRIP
3PH metallic, $k = 0.15$, α	FAIL (mem)	FAIL (mem)	FAIL (mem)	FAIL (mem)	TRIP
3PH metallic + POTT, $k = 0.1$	—	—	TRIP 32 ms	TRIP 32 ms	TRIP
Arc $R_{\text{arc}}=0.008$, $k = 0.50$, α	MISS ($R_m=0.024$)	TRIP	TRIP	TRIP	TRIP
Arc $R_{\text{arc}}=0.008$, $k = 0.15$, α	MISS	MISS ($R_m=0.061$)	MISS	MISS	TRIP
SLG metallic, $k = 0.15$, α	TRIP (67N)	TRIP (67N)	TRIP	TRIP	TRIP

FAIL (mem): Zone 2 timer $> T_{\text{mem}}$ — fixed by POTT (TR-33). MISS: $R_m > \text{reach}$ — Quad with wider R_{fwd} needed or 87L. Row 2 has no distance fix without POTT; Row 5 requires 87L for all $k_{\text{ibr}} < 0.50$.